CAN THE SEA HOLD MORE CO_2?

Over the past decade the ocean took in 29% of the CO_2 we emitted. Since 2016 that uptake has stopped growing. Ocean alkalinity enhancement (OAE) might safely raise it. Nobody knows yet. This brief asks for the independent research that would find out.

29%

of the CO_2 people emitted over the past decade was taken up by the ocean

2016

the year that uptake stopped growing, at about 3.4 billion tonnes of carbon a year

10,000+

years carbon can stay dissolved as bicarbonate, if the added alkalinity persists

WHAT IT IS

OAE adds alkaline material to surface seawater, either ground minerals such as olivine or a hydroxide made with electricity. In theory the water then holds more carbon as stable bicarbonate and draws more CO_2 from the air. It is unproven at scale and not ready to deploy.

WHAT THE EVIDENCE SHOWS

- **One permitted sea trial found no harm in what it measured.** In the Gulf of Maine in August 2025, researchers found no significant difference in plankton, fish larvae or lobster larvae. Fish, marine mammals and fisheries were not measured, and the results are not yet peer reviewed.
- **A lab result fell short.** A 2022 study found olivine removed about five times less carbon than theory predicts.
- **Verification is still an open problem.** CarbonPlan puts confidence in verifying the electrochemical route at 3 out of 5. Project Drawdown calls the evidence limited.
- **Cost and mining are open too.** Estimates are generally above $100 per tonne of CO_2, and mineral routes could need mining on the scale of the cement industry.

WHO PAYS SO FAR

- **Philanthropy.** The Carbon to Sea Initiative launched in 2023 with $50 million from foundations. It funds OAE research and also advocates for the field. It part-funded the Gulf of Maine trial, with ICONIQ Capital and NOAA.
- **Credit buyers.** In 2025 Stripe, Google, Shopify and others agreed to pay $31.3 million for 115,211 tons of future removal from one OAE company. That is not research.
- **Public money is thin.** US federal spending on all ocean methods was about $61.5 million across 2023 and 2024, then was cut in 2025. The National Academies put the need for OAE alone at $125 to $200 million. We found no oil company among the named funders.

Governments pay little while credits are on sale. That is why the ask is for public money and independent review.

FIVE ASKS FOR YOUR DELEGATION

1. **Fund independent verification research.** An open, global standard for monitoring, reporting and verification that counts the full life cycle, publicly funded and independently reviewed, in place before any credit or net zero claim counts toward a climate target.
2. **Set ecological guardrails.** Thresholds for pH, trace metals and biodiversity that pause a trial automatically, with independent oversight.
3. **Clarify governance.** Jurisdiction, liability and consent for ocean trials under UNCLOS and the London Protocol.
4. **Share decisions and benefits.** Coastal and Indigenous communities hold real decision power over trials.
5. **Keep emission cuts first.** OAE adds to rapid decarbonization and can never replace it.

TWO MINUTES WITH YOUR DELEGATION? ASK

- Does our climate research budget fund independent work on verifying OAE, and who reviews it?
- Will we keep OAE credits out of national targets until an open, public standard exists?
- Who in our delegation follows ocean carbon removal, and will they raise research governance before COP32?

Scan for every source, the full case and a letter you can send to your country's climate focal point.

alkalinityask.life

Sources: Global Carbon Budget 2025 · National Academies, 2022 · Woods Hole Oceanographic Institution, February 2026 · MIT Technology Review, 2022 and 2023 · CarbonPlan CDR Verification Framework · Project Drawdown, 2026 · Frontier, 2025 · Carbon to Sea Initiative · Pulitzer Center, 2026. Full references are on the page.

An independent brief, not affiliated with the organisations it cites. Figures as of 11 November 2026.